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Uncovering thousands of endosymbiont DNA transfer events within single cockroach genomes

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Abstract

Horizontal gene transfer (HGT) between organisms can be a valuable source of genetic variation and innovation. Research on HGT in eukaryotes has hitherto focused on transfers of coding sequences; insertions of non-coding DNA remain poorly understood. Here, we investigated HGT in cockroaches, which have a long-standing evolutionary relationship with the transovarially transmitted endosymbiont *Blattabacterium cuenoti*, making them a valuable system for assessing the potential scale of HGT. We aligned 150 bp genomic fragments of *Blattabacterium cuenoti* to 23 cockroach and termite genomes, including 8 genomes newly sequenced, and revealed pervasive endosymbiont DNA transfer events. Australian panesthiine and geoscapheine cockroaches were consistently found to harbor >3000 HGT inserts, more than an order of magnitude higher than the previous maximum estimate in other eukaryotes, excluding rotifers. Some inserts appear to have persisted for >28.7 million years in this group, which may reflect functional roles. We identified numerous chimeric inserts comprising up to nine short segments from different locations in the *Blattabacterium cuenoti* genome. Our findings indicate pervasive HGT in eukaryote genomes, with potentially far-reaching implications for adaptation and speciation.

Main Text

Introduction

The number of DNA inserts acquired via horizontal gene transfer (HGT) that become fixed in a species is expected to represent only a small fraction of the total number of transfer events that occur over evolutionary time [1]. While there is an abundant body of research on insertions with homology to protein-coding sequence, transfers that do not encode proteins, or with unknown function, have been largely ignored. Nevertheless, such insertions abruptly introduce genetic variation that can persist if fitness effects are minimal, if purifying selection is insufficient, or if the inserts acquire functional roles [2].

We explored the prevalence and characteristics of DNA inserts originating from the endosymbiont *Blattabacterium cuenoti* (hereafter *Blattabacterium*), in the genomes of 14 cockroaches and 4 termites, including 8 newly sequenced using long-read technology. *Blattabacterium* is found in specialized cells of cockroach fat bodies and is inherited transovarially [3]. Infecting the ancestors of cockroaches >150–200 Mya [4], *Blattabacterium* was lost early in termite evolution. Although long-term obligate endosymbionts have been reported to contribute no or few DNA transfers to their host genomes, previous studies focused only on protein-coding sequences [5-8].

Results and Discussion

Cockroach genomes contain thousands of Blattabacterium*-derived inserts*

By focusing on transferred DNA inserts ≥50 bp and applying stringent filters to remove artifactual inserts and inserts potentially derived from duplication events (see SI Appendix), we characterized 40,485 *Blattabacterium*-derived inserts across 23 cockroach and termite genomes (93–4900 per genome; Fig. 1A), with a median size of 161 bp (Fig. 1B). With the exception of rotifers, which have frequent HGT associated with anhydrobiosis [9], the highest number of transfers reported in any eukaryotic species was previously ~282 [8]. Inserts were clustered on some contigs, but generally were broadly distributed across host genomes, consistent with many independent HGT events (see Fig. 1C, Dataset S4 and SI Appendix).

Explanations for the high numbers of inserts we identified in cockroaches are as follows. First, we focused on *Blattabacterium*, an obligate endosymbiont whose genome is likely to come in close contact with host nuclear DNA during oogenesis and development. Second, we analyzed both intra- and intergenic regions of the genome. A recent study did not detect any horizontal transfers from *Blattabacterium* to cockroaches [8]; however, this is likely due to a focus on gene regions. Third, while previous studies filtered for the completeness of inserted coding sequences, we identified inserts as short as 50 bp. Fourth, previous studies often relied on short-read sequencing, where inserts spanning most of a sequence read may have been misidentified as bacterial contamination and discarded. In contrast, we did not actively exclude *Blattabacterium*-like sequences during assembly, and analysed genomes generated using both short- and long-read data (Dataset S2). Long-read data enables detection of inserts flanked by host sequence that may have been missed when using short-read data. Using the same pipeline, we identified comparatively low numbers of inserts from *Wolbachia*, and few to no inserts from *Buchnera* (which infects aphids, and not cockroaches) (see SI Appendix).

Panesthiinae and Geoscapheinae were found to possess particularly high numbers of *Blattabacterium*-derived inserts (Fig. 1A). Previous work found that this group has experienced considerable genome-wide relaxed selection [10], possibly due to reduced effective population sizes, which may allow larger numbers of inserts to persist. The low number of inserts identified in termite genomes may reflect the long-term absence of *Blattabacterium*, as well as a considerable reduction in genome size compared with cockroaches.

Evolution of Blattabacterium*-derived inserts in cockroach genomes*

Examination of insert position across the cockroach genomes revealed that 15.38–25.56% were found in intragenic regions, and 74.44–84.62% were found in intergenic regions. This differs somewhat from the proportion of genic regions in the genomes, which was >25% in all but four (range 13.79–33.71%). The average size of inserts (+/- standard deviation) in intragenic regions (199.13 +/- 138.75 bp) was similar to that in intergenic regions (191.70 +/- 126.24 bp), although significantly higher in the former (one-way nested analysis of variance: $F_{1,39424} = 17.96$, $p < 0.01$; Fig. 1B). Insert sequences had GC content (17.9–29.6%) similar to those of *Blattabacterium* genomes (21.3–28.2%), but considerably lower than those of cockroach and termite genomes (34.5–41.1%). There was no apparent bias in the origin of inserts with respect to their position in the *Blattabacterium* genome, which is highly conserved in size, gene content, and gene order across strains (Fig. 1D).

Closely related taxa carried similar numbers of inserts, suggesting that many had been inherited vertically (Fig. 1A). We identified putatively ancestral HGT inserts by calculating their similarity to inserts in other species and to the host's specific *Blattabacterium* strain, and subsequently inspecting homology (and common origin) of cockroach genome sequence upstream and downstream from these candidate ancestral inserts (n = 22) across multiple taxa (Fig. 2A; see SI Appendix for details). The oldest ancestral inserts identified were inferred to have been present in the most recent common ancestor of all Panesthiinae and Geoscapheinae, at least ~28.7 Mya [11].

Transcription and chimeric nature of inserts

We used RNAseq data to investigate the transcription of inserts in *Panesthia t. tegminifera*, *Macropanesthia lithgowae*, and *Panesthia sloanei*. By mapping insert sequences to RNA contigs using BLASTn, we found that 91.42–94.96% of inserts showed no evidence of transcription (although RNA from all life stages and tissues was not assessed). Of the remaining inserts, which were observed in assembled RNA contigs, 26.59–32.97% overlapped with annotated genic regions of the cockroach genomes (1.66–2.28% of total inserts), while the remainder overlapped with intergenic regions (3.38–6.30% of total inserts). A small fraction (0.40–0.65%) of inserts mapped to exon sequences within RNA contigs (see Dataset S5 for gene details).

During BLAST inspections of the longest inserts, we identified 32 inserts displaying a chimeric nature, comprising *Blattabacterium*-like sequences from distant areas within the *Blattabacterium* genome placed directly adjacent to each other within host genome contigs (Fig. 2B). These chimeric inserts were present in all cockroach genomes, comprising up to nine *Blattabacterium*-derived segments. Similar to chimeric numts previously identified in *Arabidopsis thaliana* [12], they are likely to have been produced by non-homologous end-joining repair that incorporated segments of *Blattabacterium* DNA that possibly arose following symbiont cell replication or death.

Conclusion

Our findings reveal extensive horizontal transfer of DNA from prokaryote symbionts to eukaryotes. Furthermore, the persistence of numerous inserts over millions of years indicates that they may have assumed functional roles in both genes and intergenic regions, are effectively neutral, or are only slightly deleterious. Future research on cockroaches and other species harboring obligate endosymbiotic prokaryotes will help to uncover any functional effects of inserts, providing a more comprehensive understanding of how HGT shapes genome evolution.

Materials and Methods

Detailed methods are described in the SI Appendix and on GitHub (https://github.com/MEEP-projects/HGT_pipeline). Briefly, we generated 8 new genome assemblies from species of Blaberidae (Dataset S1), and analyzed these with 15 previously generated genomes from Blattodea and 77 *Blattabacterium* (Dataset S2). To locate DNA transfers from *Blattabacterium* to host genomes, we aligned 150-bp fragments of the *Blattabacterium* genomes to the cockroach and termite genomes. Aligned fragments were characterized as putative HGT inserts, with a minimum length of 50 bp. Several filtering steps and QC analyses were applied to confirm that the identified inserts were ‘true’ HGT inserts, rather than assembly artifacts or misidentified *Blattabacterium* DNA, and to identify unique insert events within a genome (i.e., excluding inserts arising from duplication or transposition). We also ran control analyses using *Buchnera* (negative control) and *Wolbachia* (known to infect hosts) using the same pipeline. After characterizing inserts, we investigated their minimum age, length, origin, insertion locality, and GC content.

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Figures

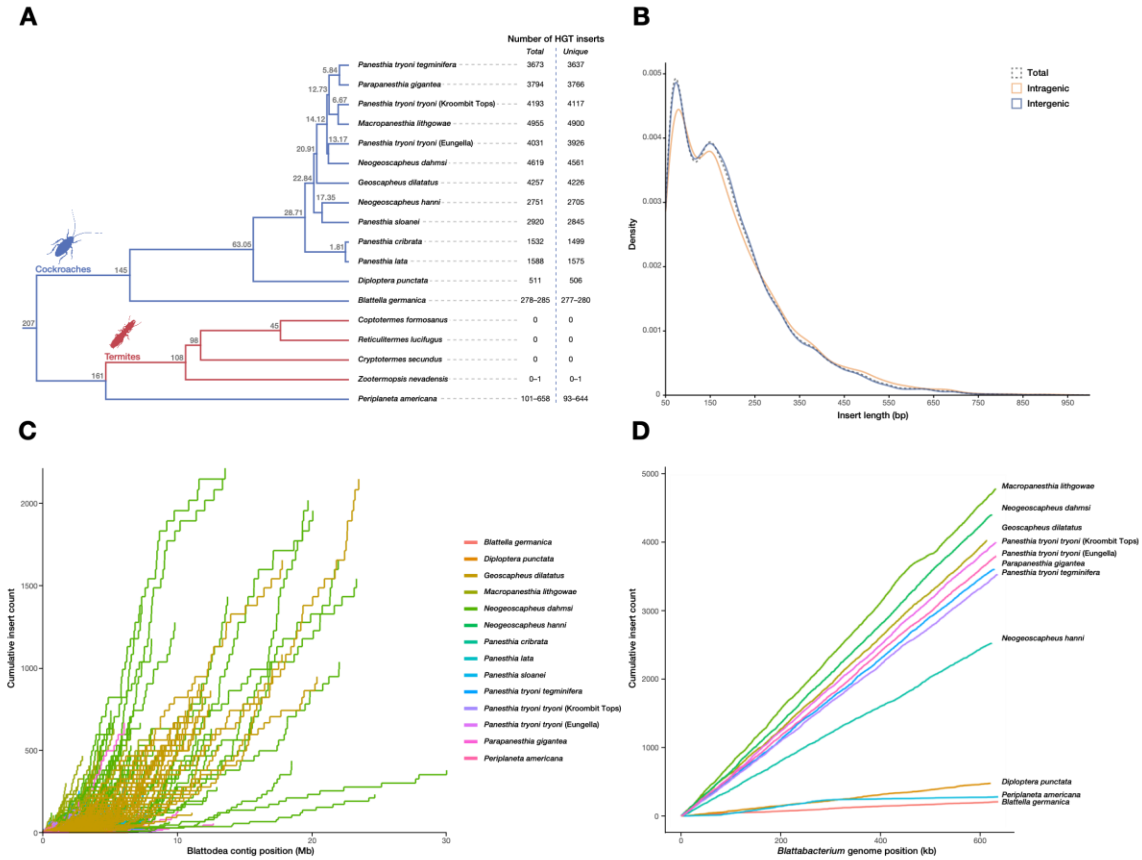

Figure 1. Dated evolutionary tree of analyzed taxa alongside total number of HGT inserts and number of unique inserts (excluding potential cases of duplication or transposition) (A). Ranges are given for taxa with multiple analyzed genomes. Node ages (Mya) are based on previous studies [11,13,14]. Length distribution of inserts (B). 50 bp represents our minimum length threshold, and inserts >1000 bp are omitted for clarity (51 inserts omitted). Cumulative number of inserts plotted along host genomes (C). Individual lines correspond to individual contigs (coloured by genome), such that each line starts at position 0 and terminates at the length of the contig. Cumulative number of inserts plotted along the *Blattabacterium* genome (D), based on the start position of each insert. Linear trends indicating random origins from the *Blattabacterium* genome. Only species with a published, host-specific *Blattabacterium* strain are included (Dataset S2). For species with multiple genomes, the version with the most inserts was analyzed (for B-D).

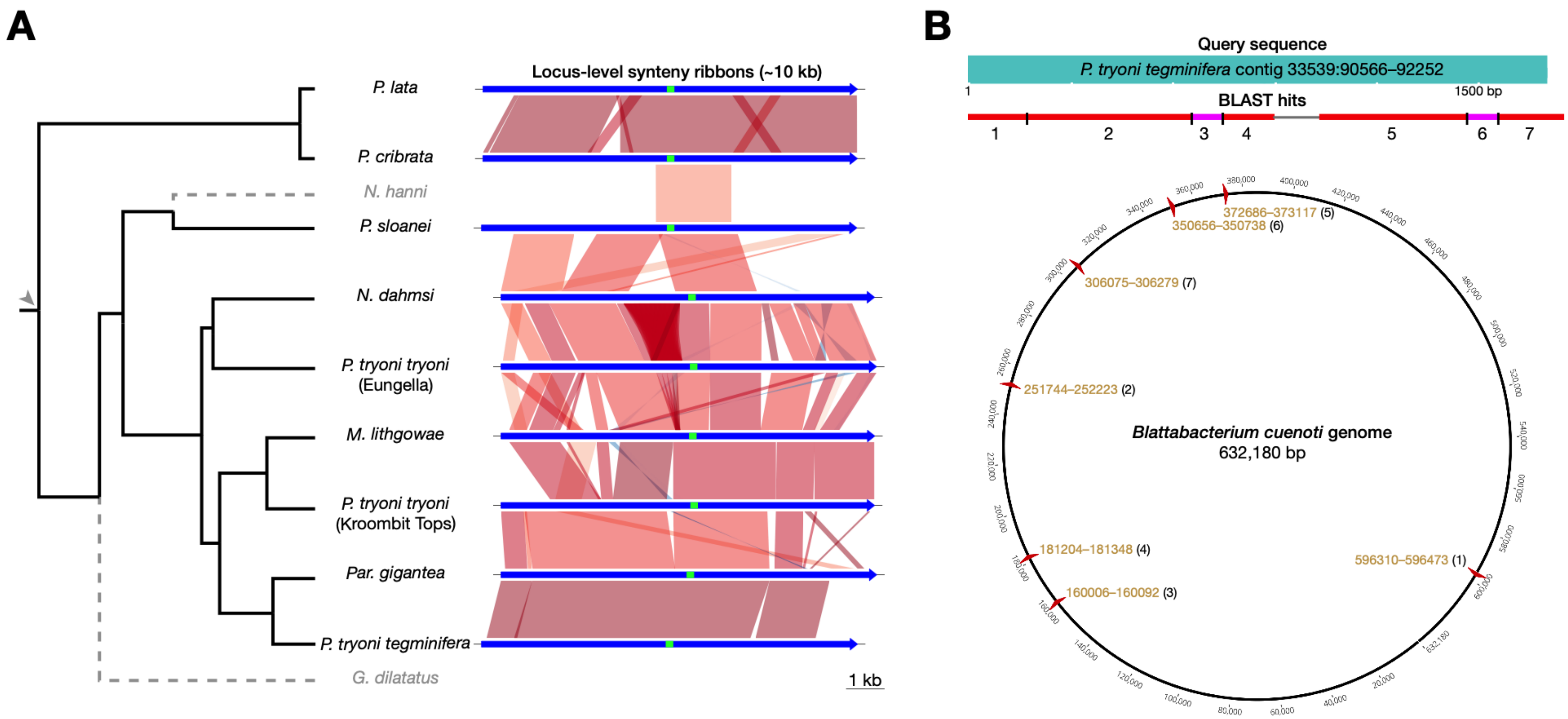

Figure 2. Putative ancestral insert evaluated by examination of flanking region homology (A). Homology of an insert (green bar) and its 5-kb flanking regions (blue bars) across species is shown on right (species lacking the insertion are shown with dashed lines on left). Arrowhead denotes inferred ancestral insertion event. Example of a chimeric HGT insertion in the *Panesthia tryoni tegminifera* genome (B). The *P. t. tegminifera* contig (top) is divided into seven segments of different origin, and the circular *Blattabacterium cuenoti* genome (CP142624.1; bottom) indicates the source of these insertions. Numbers indicate corresponding segments in the cockroach and *Blattabacterium cuenoti* genomes (with yellow labels marking genomic coordinates).


A
Number of HGT inserts
Total
Unique
Panesthia tryoni tegminifera 3673 3637
Parapanesthia gigantea 3794 3766
Panesthia tryoni tryoni (Kroombit Tops) 4193 4117
Macropanesthia lithgowae 4955 4900
Panesthia tryoni tryoni (Eungella) 4031 3926
Neogeoscapheus dahmsi 4619 4561
Geoscapheus dilatatus 4257 4226
Neogeoscapheus hanni 2751 2705
Panesthia sloanei 2920 2845
Panesthia cribrata 1532 1499
Panesthia lata 1588 1575
Diploptera punctata 511 506
Blattella germanica 278–285 277–280
Coptotermes formosanus 0 0
Reticulitermes lucifugus 0 0
Cryptotermes secundus 0 0
Zootermopsis nevadensis 0–1 0–1
Periplaneta americana 101–658 93–644
5.84
12.73
6.67
14.12
13.17
20.91
22.84
17.35
28.71
1.81
63.05
145
207
45
98
108
161
Cockroaches
Termites
B
Total
Intragenic
Intergenic
Density
Insert length (bp)
0.005
0.004
0.003
0.002
0.001
0
50
150
250
350
450
550
650
750
850
950
C
Cumulative insert count
Blattodea contig position (Mb)
2000
1500
1000
500
0
10
20
30
Blattella germanica
Diploptera punctata
Geoscapheus dilatatus
Macropanesthia lithgowae
Neogeoscapheus dahmsi
Neogeoscapheus hanni
Panesthia cribrata
Panesthia lata
Panesthia sloanei
Panesthia tryoni tegminifera
Panesthia tryoni tryoni (Kroombit Tops)
Panesthia tryoni tryoni (Eungella)
Parapanesthia gigantea
Periplaneta americana
D
Cumulative insert count
Blattabacterium genome position (kb)
5000
4000
3000
2000
1000
0
200
400
600
Macropanesthia lithgowae
Neogeoscapheus dahmsi
Geoscapheus dilatatus
Panesthia tryoni tryoni (Kroombit Tops)
Panesthia tryoni tryoni (Eungella)
Parapanesthia gigantea
Panesthia tryoni tegminifera
Neogeoscapheus hanni
Diploptera punctata
Periplaneta americana
Blattella germanica


A
Locus-level synteny ribbons (~10 kb)
P. lata
P. cribrata
N. hanni
P. sloanei
N. dahmsi
P. tryoni tryoni (Eungella)
M. lithgowae
P. tryoni tryoni (Kroombit Tops)
Par. gigantea
P. tryoni tegminifera
G. dilatatus
1 kb
B
Query sequence
P. tryoni tegminifera contig 33539:90566–92252
1
1500 bp
BLAST hits
1 2 3 4 5 6 7
372686–373117 (5)
350656–350738 (6)
306075–306279 (7)
251744–252223 (2)
181204–181348 (4)
160006–160092 (3)
596310–596473 (1)
Blattabacterium cuenoti genome
632,180 bp